II. Public Debt Sustainability Analysis

7. **The baseline fiscal framework is consistent with lower total public debt and comfortable debt-service ratios.** The NPV of the public debt falls from 51.1 to 39.2 percent of GDP over the twenty-year projection horizon. A decline in the domestic debt by 15.4 percent of GDP is partly offset by an increase in the NPV of external liabilities by about 3.5 percent of GDP. Debt-service ratios absorb an increasing share of revenue in line with the gradual increase in projected real borrowing rates, and the high initial level of domestic debt at 33 percent of GDP. However, debt service remains at manageable levels.

8. **The public debt DSA demonstrates that the principal risks to the scenario include lower average growth and elevated primary deficits** as described below:

- **The main risk to public debt sustainability is permanently lower GDP growth** by one standard deviation relative to the baseline path. This alternative scenario would result in a higher NPV of public debt by about 25 percent of GDP compared to 2005/06. Moreover, debt service would also reach 13 percent of revenue in 2025/26 compared to just 8 percent in the baseline.
- **Debt sustainability prospects would also be adversely affected if average growth and the primary deficit remain at historical levels.** In this scenario, the NPV of the public debt increases by 7 percent of GDP compared to 2005/06, and debt service reaches 11 percent of revenue.
- **The bound tests confirm the sensitivity of public debt sustainability to lower growth.** The NPV of public debt as a share of GDP decreases by 2025/26 in each bound test except the case of a transitory shock in real GDP growth during 2006/07 and 2007/08 .[9] In this case, the NPV of public debt as a share of GDP peaks at 75 percent in 2015/16 after increasing 24 percent of GDP, and debt service reaches 13 percent of revenue compared to 8 percent in the baseline. This example highlights the challenge of maintaining a sustainable public debt given Ethiopia's vulnerability to severe climatic shocks that result in volatile growth swings.

9. **The results of the public DSA reiterate the importance of developing an comprehensive public debt strategy.** The strategy should develop separate targets and adjustment paths for the domestic and external debt based on a range of fiscal indicators. In this manner, medium-term budget planning should be anchored by the public debt strategy. Contingency plans should play an integral role in the strategy to ensure that the public debt remains sustainable following severe shocks. In this context, lowering the domestic debt should be a priority to provide scope for higher borrowing to absorb the fiscal impact of macroeconomic shocks.

[9] The bound test for lower real GDP growth is calculated by reducing projected growth by one standard deviation in 2006/07 and 2007/08. This corresponds to a contraction in real GDP by 0.9 percent compared to the baseline growth projection of 5.7 percent.

III. Impact of the full delivery of debt relief under the MDRI

10. **Delivery of MDRI debt relief from IDA and the AfDB would improve Ethiopia's debt sustainability outlook by leading to a sharp reduction of Ethiopia's debt burden indicators** (text table).[10] The NPV of debt-to export ratio would decrease by about 50 percentage points in the initial year compared to the baseline, and would remain comfortably below the indicative threshold throughout the projection period. Similarly, the NPV of debt-to-GDP would decrease by 6 percentage points on average compared to the baseline. Under the full MDRI scenario, the debt service would average 5.3 percent, increasing from 3.9 percent in 2006/07 to 7.7 percent in 2012/13 to then decline to about 4.0 percent by the end of the projection period. Albeit the sharp initial impact of MDRI debt relief, all debt burden indicators converge over time to the pre-MDRI situation, with the NPV of debt-to-GDP ratio at the end of the projection period above its initial levels.

Table 2. Ethiopia: Impact of the full provision of MDRI on the baseline scenario

	2005/06	2009/10	2012/13	2015/16	2025/26
		NPV of debt-to-GDP ratio			
Before MDRI 1/	17.6	25.7	29.2	27.7	21.5
After MDRI	9.0	18.1	22.5	21.7	19.2
		NPV of exports-to-GDP ratio			
Before MDRI 1/	102.2	132.3	144.0	130.1	78.6
After MDRI	51.9	93.0	110.9	102.2	70.5
		Debt service ratio			
Before MDRI 1/	5.2	5.2	8.2	6.9	5.0
After MDRI	5.2	4.7	7.7	6.5	3.9

1/ Includes the IMF debt relief under the MDRI.

11. **Under the current CPIA classification, and contingent on a strong economic performance, a reassessment of Ethiopia's risk of debt distress could be needed once MDRI from IDA and the AfDB is approved.**

[10] This alternative baseline assumes that MDRI debt relief from IDA is implemented starting in July 2006 through the forgiveness of the debt service payments on the debt outstanding and disbursed (DOD) as of end-December 2003. As for the African Development Fund, it is assumed that debt service payments on the DOD as of end-December 2004 with retroactive application to January-2006. Under those assumptions, IDA and the AfDB would deliver MDRI debt relief in the amount of approximately SDR1.6 billion (equivalent to US$2.3 billion at the exchange rate prevailing at end-December 2005) and US$0.9 billion, respectively.

Table 1a. Ethiopia: External Debt Sustainability Framework, Baseline Scenario, 2002-2025 1/
(In percent of GDP, unless otherwise indicated)

	Actual		Estimate	Historical Average 6/	Standard Deviation 6/	Projections											
	2002/03	2003/04	2004/05			2005/06	2006/07	2007/08	2008/09	2009/10	2012/13	2013/14	2014/15	2015/16	2016/17	2024/25	2025/26
External debt (nominal) 1/	**85.4**	**75.7**	**53.9**			**47.4**	**48.1**	**47.5**	**46.0**	**45.3**	**42.6**	**41.6**	**40.7**	**39.9**	**39.1**	**31.1**	**30.0**
o/w public and publicly guaranteed (PPG)	85.4	75.7	53.9			47.4	48.1	47.5	46.0	45.3	42.6	41.6	40.7	39.9	39.1	31.1	30.0
Change in external debt	1.9	-9.7	-21.8			-6.5	0.7	-0.6	-1.5	-0.7	-1.1	-1.0	-0.9	-0.8	-0.8	-1.0	-1.0
Identified net debt-creating flows	-5.6	-11.6	-2.0			3.7	0.5	-0.9	-0.1	0.3	-0.8	-2.2	-2.8	-2.8	-2.7	-1.2	-1.6
Non-interest current account deficit	**1.3**	**4.5**	**8.6**	**2.9**	**3.4**	**7.0**	**3.8**	**2.5**	**3.1**	**3.4**	**1.7**	**0.2**	**-0.4**	**-0.5**	**-0.5**	**0.5**	**0.1**
Deficit in balance of goods and services	15.2	17.2	22.7			20.4	19.1	18.7	19.3	19.1	17.0	15.4	14.6	14.4	14.3	15.8	15.9
Exports	14.4	15.4	16.4			17.3	18.4	19.0	19.2	19.4	20.3	20.6	20.9	21.3	21.7	26.5	27.3
Imports	29.5	32.6	39.1			37.7	37.5	37.7	38.5	38.5	37.3	36.0	35.5	35.6	36.0	42.3	43.2
Net current transfers (negative = inflow)	-13.8	-12.7	-14.0	-9.8	2.8	-13.1	-14.9	-15.8	-15.8	-15.3	-14.7	-14.4	-14.1	-13.9	-13.7	-14.2	-14.8
Other current account flows (negative = net inflow)	-0.1	0.0	-0.1			-0.3	-0.3	-0.4	-0.3	-0.4	-0.6	-0.8	-0.9	-1.0	-1.0	-1.1	-1.1
Net FDI (negative = inflow)	**-1.4**	**-1.0**	**-1.3**	**-1.0**	**0.7**	**-1.3**	**-1.3**	**-1.3**	**-1.3**	**-1.2**	**-1.0**	**-1.0**	**-1.0**	**-0.9**	**-0.8**	**-0.5**	**-0.5**
Endogenous debt dynamics 2/	**-5.5**	**-15.1**	**-9.3**			**-2.0**	**-2.0**	**-2.0**	**-2.0**	**-1.8**	**-1.5**	**-1.4**	**-1.4**	**-1.4**	**-1.4**	**-1.2**	**-1.2**
Contribution from nominal interest rate	0.9	0.6	0.5			0.5	0.4	0.4	0.4	0.5	0.7	0.7	0.7	0.6	0.6	0.4	0.4
Contribution from real GDP growth	2.5	-7.7	-5.8			-2.4	-2.4	-2.5	-2.4	-2.3	-2.2	-2.1	-2.1	-2.0	-2.0	-1.6	-1.6
Contribution from price and exchange rate changes	-8.9	-8.0	-4.0			…	…	…	…	…	…	…	…	…	…	…	…
Residual (3-4) 3/	**7.5**	**2.0**	**-19.8**			**-10.2**	**0.2**	**0.3**	**-1.4**	**-1.0**	**-0.3**	**1.2**	**1.9**	**2.0**	**1.9**	**0.2**	**0.6**
o/w exceptional financing	-0.9	-0.8	-0.8			0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
NPV of external debt 4/	…	…	17.6			17.6	19.3	20.7	23.5	25.7	29.2	28.6	28.1	27.7	27.2	22.2	21.5
In percent of exports	…	…	107.8			102.2	105.1	108.8	122.4	132.3	144.0	139.0	134.4	130.1	125.6	83.8	78.6
NPV of PPG external debt	**…**	**…**	**17.6**			**17.6**	**19.3**	**20.7**	**23.5**	**25.7**	**29.2**	**28.6**	**28.1**	**27.7**	**27.2**	**22.2**	**21.5**
In percent of exports	…	…	107.8			102.2	105.1	108.8	122.4	132.3	144.0	139.0	134.4	130.1	125.6	83.8	78.6
Debt service-to-exports ratio (in percent)	15.2	12.3	7.1			5.2	4.4	4.2	4.0	5.2	8.2	8.1	7.3	6.9	6.7	5.2	5.0
PPG debt service-to-exports ratio (in percent)	15.2	12.3	7.1			5.2	4.4	4.2	4.0	5.2	8.2	8.1	7.3	6.9	6.7	5.2	5.0
Total gross financing need (billions of U.S. dollars)	0.2	0.5	0.9			0.8	0.4	0.3	0.4	0.5	0.5	0.2	0.0	0.0	0.0	0.7	0.5
Non-interest current account deficit that stabilizes debt ratio	-0.6	14.2	30.5			13.5	3.1	3.1	4.7	4.1	2.8	1.2	0.5	0.3	0.4	1.5	1.1
Key macroeconomic assumptions																	
Real GDP growth (in percent)	-3.3	11.1	8.8	4.7	5.9	5.2	5.4	5.5	5.5	5.5	5.4	5.4	5.4	5.3	5.4	5.4	5.4
GDP deflator in US dollar terms (change in percent)	12.0	10.3	5.5	-1.0	14.1	8.8	-0.1	1.7	2.4	2.2	2.4	2.4	2.3	2.0	1.9	1.7	1.7
Effective interest rate (percent) 5/	1.2	0.9	0.7	1.2	0.3	1.0	0.9	1.0	1.0	1.3	1.8	1.9	1.7	1.6	1.5	1.2	1.2
Growth of exports of G&S (US dollar terms, in percent)	16.0	31.5	22.1	9.7	14.3	20.7	12.3	10.8	9.1	9.0	9.4	9.5	9.5	9.2	9.3	10.4	10.5
Growth of imports of G&S (US dollar terms, in percent)	13.2	35.1	37.7	13.8	13.0	10.3	4.7	7.9	10.3	8.0	8.6	4.3	6.2	7.8	8.4	9.7	9.6
Grant element of new public sector borrowing (in percent)	…	…	…	…	…	41.8	47.6	49.6	37.4	35.4	35.8	49.7	49.9	50.0	50.2	51.6	51.8
Memorandym item:																	
Nominal GDP (billions of US dollars)	7.9	9.7	11.2			12.8	13.5	14.5	15.6	16.8	21.2	22.8	24.6	26.4	28.4	49.1	52.6

Source: Staff simulations.

1/ Includes both public and private sector external debt.
2/ Derived as $[r - g - \rho(1+g)]/(1+g+\rho+g\rho)$ times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that NPV of private sector debt is equivalent to its face value.
5/ Current-year interest payments devided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

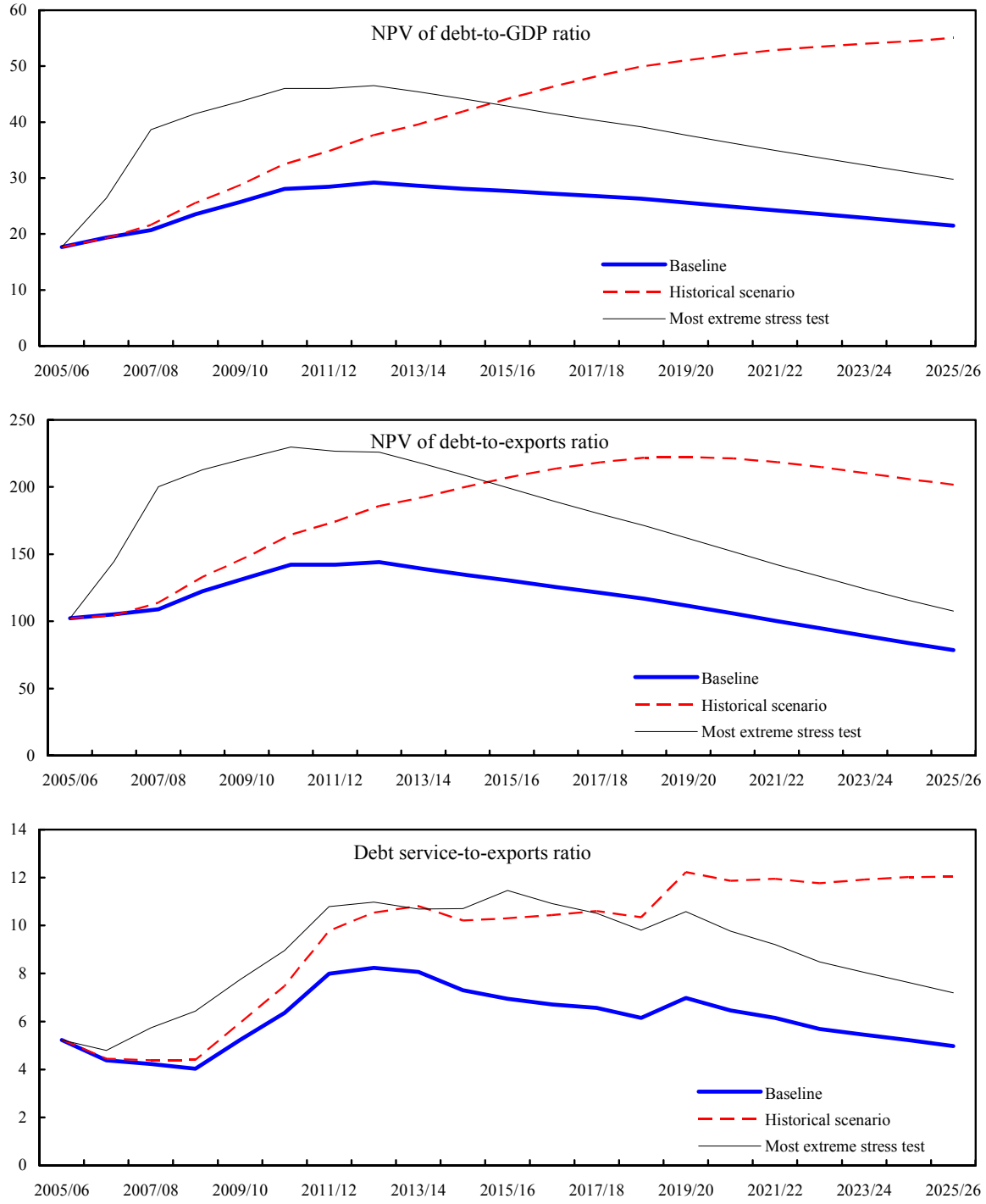
Table 1b. Ethiopia: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2005-25
(In percent)

	Estimate						Projections							
	2005/06	2006/07	2007/08	2008/09	2009/10	2010/11	2011/12	2012/13	2013/14	2014/15	2015/16	2016/17	2024/25	2025/26
NPV of debt-to-GDP ratio														
Baseline	17.6	19.3	20.7	23.5	25.7	28.1	28.4	29.2	28.6	28.1	27.7	27.2	22.2	21.5
A. Alternative Scenarios														
A1. Key variables at their historical averages in 2006-25 1/	17.6	19.2	21.6	25.5	28.7	32.5	34.8	37.7	39.6	41.9	44.2	46.3	54.4	55.0
A2. New public sector loans on less favorable terms in 2006-25 2/	17.6	20.6	23.2	28.1	32.0	36.2	37.7	39.8	39.8	39.9	39.9	39.9	36.5	35.8
B. Bound Tests														
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	17.6	20.6	23.5	26.7	29.3	31.9	32.4	33.2	32.6	32.0	31.5	31.0	25.2	24.4
B2. Export value growth at historical average minus one standard deviation in 2006-07 3/	17.6	20.9	25.1	27.7	29.8	32.0	32.2	32.8	32.1	31.4	30.7	30.0	23.6	22.7
B3. US dollar GDP deflator at historical average minus one standard deviation in 2006-07	17.6	21.1	25.1	28.5	31.2	34.1	34.5	35.5	34.7	34.1	33.6	33.0	26.9	26.0
B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 4/	17.6	24.4	31.2	33.6	35.4	37.4	37.4	37.9	37.0	36.0	34.9	33.9	25.5	24.5
B5. Combination of B1-B4 using one-half standard deviation shocks	17.6	26.4	38.7	41.5	43.7	46.0	46.0	46.5	45.4	44.2	42.9	41.6	31.1	29.8
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 5/	17.6	27.7	29.6	33.6	36.7	40.1	40.6	41.7	40.9	40.1	39.5	38.9	31.7	30.7
NPV of debt-to-exports ratio														
Baseline	102.2	105.1	108.8	122.4	132.3	142.3	142.2	144.0	139.0	134.4	130.1	125.6	83.8	78.6
A. Alternative Scenarios														
A1. Key variables at their historical averages in 2006-25 1/	102.2	104.6	113.8	132.9	147.8	164.6	174.2	185.8	192.4	200.5	207.7	213.7	205.8	201.7
A2. New public sector loans on less favorable terms in 2006-25 2/	102.2	111.9	122.2	146.2	164.8	183.4	188.6	196.1	193.7	190.9	187.8	184.1	138.0	131.4
B. Bound Tests														
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	102.2	105.1	108.8	122.4	132.3	142.3	142.2	144.0	139.0	134.4	130.1	125.6	83.8	78.6
B2. Export value growth at historical average minus one standard deviation in 2006-07 3/	102.2	133.8	180.4	197.4	209.6	221.7	220.3	221.5	213.4	205.6	197.7	189.6	121.9	114.0
B3. US dollar GDP deflator at historical average minus one standard deviation in 2006-07	102.2	105.1	108.8	122.4	132.3	142.3	142.2	144.0	139.0	134.4	130.1	125.6	83.8	78.6
B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 4/	102.2	132.6	163.9	174.8	182.3	189.6	187.2	186.8	179.7	172.2	164.4	156.6	96.4	89.7
B5. Combination of B1-B4 using one-half standard deviation shocks	102.2	144.7	200.3	212.9	221.5	229.8	226.7	225.9	217.3	208.2	198.5	188.9	115.6	107.5
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 5/	102.2	105.1	108.8	122.4	132.3	142.3	142.2	144.0	139.0	134.4	130.1	125.6	83.8	78.6
Debt service ratio														
Baseline	5.2	4.4	4.2	4.0	5.2	6.4	8.0	8.2	8.1	7.3	6.9	6.7	5.2	5.0
A. Alternative Scenarios														
A1. Key variables at their historical averages in 2006-25 1/	5.2	4.4	4.4	4.4	5.9	7.5	9.8	10.5	10.8	10.2	10.3	10.4	12.0	12.0
A2. New public sector loans on less favorable terms in 2006-25 2/	5.2	4.4	4.4	4.7	5.7	6.3	7.4	7.8	8.0	8.3	10.2	9.7	9.2	8.8
B. Bound Tests														
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	5.2	4.4	4.2	4.0	5.2	6.4	8.0	8.2	8.1	7.3	6.9	6.7	5.2	5.0
B2. Export value growth at historical average minus one standard deviation in 2006-07 3/	5.2	5.2	6.0	6.2	7.8	9.3	11.5	11.8	11.5	10.8	10.9	10.5	7.8	7.4
B3. US dollar GDP deflator at historical average minus one standard deviation in 2006-07	5.2	4.4	4.2	4.0	5.2	6.4	8.0	8.2	8.1	7.3	6.9	6.7	5.2	5.0
B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 4/	5.2	4.4	4.9	5.3	6.4	7.5	9.0	9.2	8.9	9.0	9.4	9.0	6.3	6.0
B5. Combination of B1-B4 using one-half standard deviation shocks	5.2	4.8	5.7	6.4	7.7	9.0	10.8	11.0	10.7	10.7	11.5	10.9	7.6	7.2
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 5/	5.2	4.4	4.2	4.0	5.2	6.4	8.0	8.2	8.1	7.3	6.9	6.7	5.2	5.0
Memorandum item:														
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	43.7	43.7	43.7	43.7	43.7	43.7	43.7	43.7	43.7	43.7	43.7	43.7	43.7	43.7

Source: Staff projections and simulations.

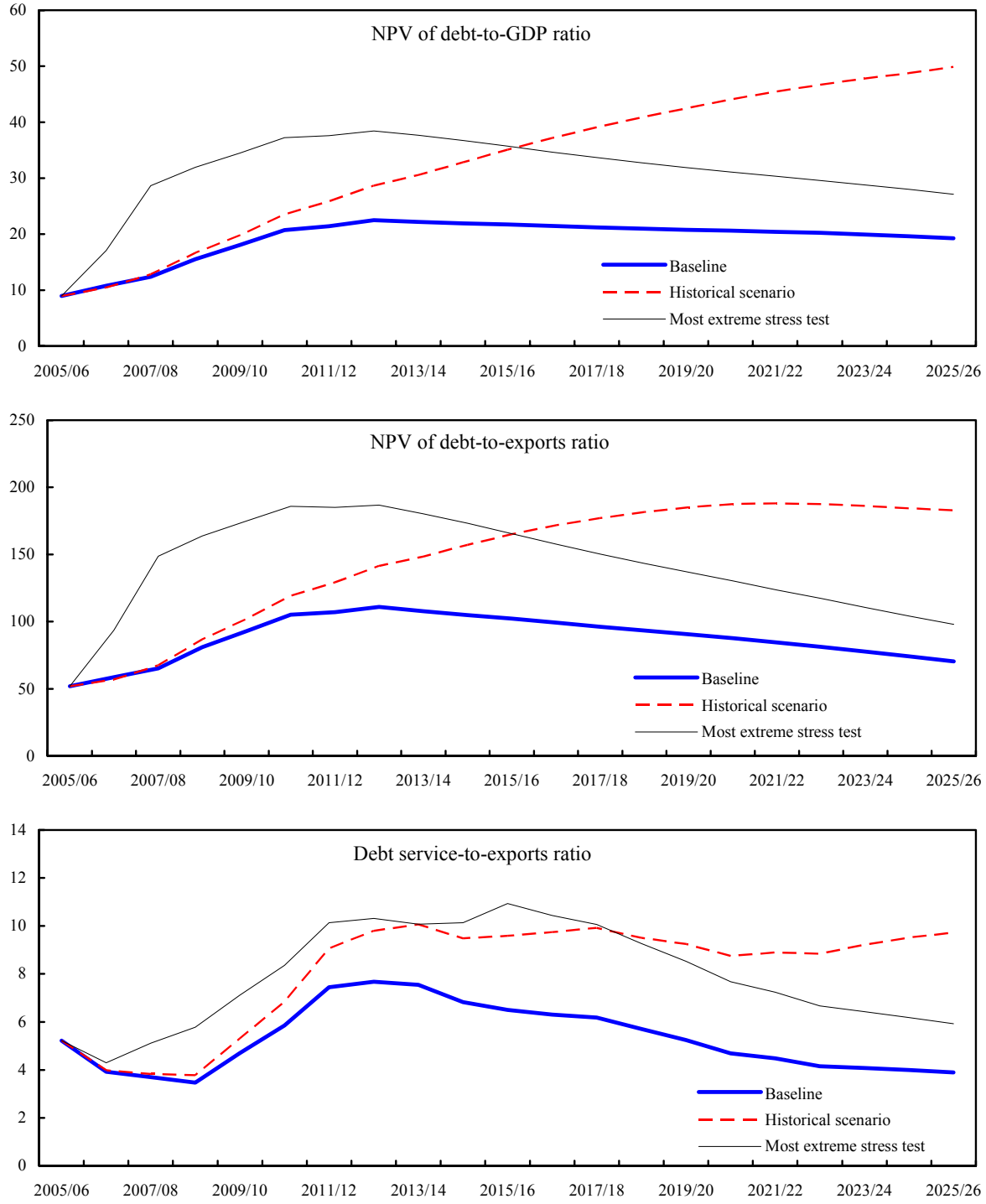
1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Figure 2a. Ethiopia: Indicators of Public and Publicly Guaranteed External Debt
Under Alternative Scenarios, 2005/06-2025/26
(In percent)



Source: Staff projections and simulations.

Figure 2b. Ethiopia: Indicators of Public and Publicly Guaranteed External Debt
Under Alternative Scenarios After full MDRI, 2005/06-2025/26
(In percent)



Source: Staff projections and simulations.

Table 2a.Ethiopia: Public Sector Debt Sustainability Framework, Baseline Scenario, 2002/3-2025/26
(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2002/3	2003/4	2004/5	Historical Average 5/	Standard Deviation 5/	2005/6	2006/7	2007/8	2008/9	2009/10	2010/11	2005/6-2010/11 Average	2015/16	2025/26	2011/12-2025/26 Average
Public sector debt 1/	124.5	110.9	88.8			82.0	81.9	81.1	79.0	77.8	76.1		67.3	47.9	
o/w foreign-currency denominated	85.6	75.9	54.0			49.2	49.1	48.4	46.9	46.2	45.4		40.7	30.6	
Change in public sector debt	1.3	-13.6	-22.1			-6.8	-0.1	-0.8	-2.1	-1.1	-1.7		-1.6	-1.9	
Identified debt-creating flows	-4.2	-20.6	-10.8			-5.9	-1.9	-2.4	-1.9	-1.5	-2.1		-1.6	-1.1	
Primary deficit	5.2	1.8	3.9	4.0	2.4	4.0	4.5	3.7	3.6	3.5	3.0	3.7	2.2	1.5	1.9
Revenue and grants	23.0	21.4	20.7			23.5	22.1	21.9	22.8	23.2	23.8		24.4	24.6	
of which : grants	6.7	4.8	4.7			5.4	5.3	5.4	6.3	6.7	7.2		7.4	7.2	
Primary (noninterest) expenditure	28.3	23.2	24.6			27.5	26.5	25.5	26.3	26.6	26.8		26.6	26.1	
Automatic debt dynamics	-8.6	-21.6	-13.6			-9.0	-5.5	-5.2	-5.4	-5.0	-5.1		-3.8	-2.6	
Contribution from interest rate/growth differential	-0.4	-16.8	-11.5			-7.7	-5.6	-5.5	-5.6	-5.1	-5.0		-3.8	-2.7	
of which : contribution from average real interest rate	-4.3	-3.2	-2.6			-3.2	-1.3	-1.0	-1.2	-0.9	-1.0		-0.4	-0.1	
of which : contribution from real GDP growth	3.9	-13.6	-8.9			-4.5	-4.4	-4.4	-4.4	-4.2	-4.1		-3.5	-2.6	
Contribution from real exchange rate depreciation	-8.3	-4.8	-2.1			-1.4	0.1	0.3	0.1	0.2	-0.1		...	...	
Other identified debt-creating flows	-0.8	-0.8	-1.0			-0.9	-0.9	-0.9	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	-0.2			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other) 6/	-0.8	-0.8	-0.8			-0.9	-0.9	-0.9	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	5.5	6.9	-11.3			-0.9	1.9	1.6	-0.3	0.4	0.4		-0.1	-0.8	
NPV of public sector debt	39.0	35.0	52.5			51.1	52.6	53.8	56.1	57.9	59.3		54.8	39.2	
o/w foreign-currency denominated	13.0	12.6	17.7			18.3	19.7	21.1	23.9	26.2	28.6		28.2	21.9	
Gross financing need 2/	22.1	22.6	23.6			24.5	26.1	26.6	26.3	26.8	26.7		25.4	18.5	
NPV of public sector debt-to-revenue ratio (in percent) 3/	169.1	163.9	253.3			217.0	238.1	246.0	246.3	249.7	249.7		224.9	159.2	
o/w external	...	...	85.3			77.9	89.4	96.4	105.2	113.0	120.4		115.6	88.9	
Debt service-to-revenue ratio (in percent) 3/ 4/	11.3	10.7	9.3			8.8	9.8	10.5	6.6	7.1	7.3		8.5	7.9	
Primary deficit that stabilizes the debt-to-GDP ratio	3.9	15.4	26.0			10.8	4.5	4.5	5.7	4.6	4.8		3.9	3.4	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	-3.1	12.3	8.7	5.0	5.9	5.3	5.7	5.7	5.7	5.7	5.5	5.6	5.3	5.4	5.4
Average nominal interest rate on forex debt (in percent)	1.1	0.9	0.8	0.9	0.2	1.0	1.0	0.9	0.5	0.5	0.5	0.7	0.5	0.6	0.6
Average real interest rate on domestic currency debt (in percent)	-9.3	-6.6	-4.2	-1.0	7.3	-8.2	-2.6	-1.7	-1.5	-0.9	-1.1	-2.7	0.8	1.6	1.0
Real exchange rate depreciation (in percent, + indicates depreciation)	-9.6	-6.3	-3.1	3.1	8.2	-2.7	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	13.0	9.4	6.2	2.5	5.9	11.6	5.8	5.3	5.7	5.5	6.2	6.7	5.9	5.6	5.9
Growth of real primary spending (deflated by GDP deflator, in percent)	2.5	-7.9	15.1	10.4	13.9	18.0	1.6	1.9	9.5	7.4	6.4	7.5	4.2	5.3	5.2
Grant element of new external borrowing (in percent)	...	...	...	...	...	41.8	47.6	49.6	37.4	35.4	30.1	40.3	50.0	51.8	...

Sources: Country authorities; and Fund staff estimates and projections.

1/ Public sector debt corresponds to the general government gross debt, including publicly guaranteed debt to the public enterprises and the external debt of Ethiopia Airlines. The fiscal year ends on July 7.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues including grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

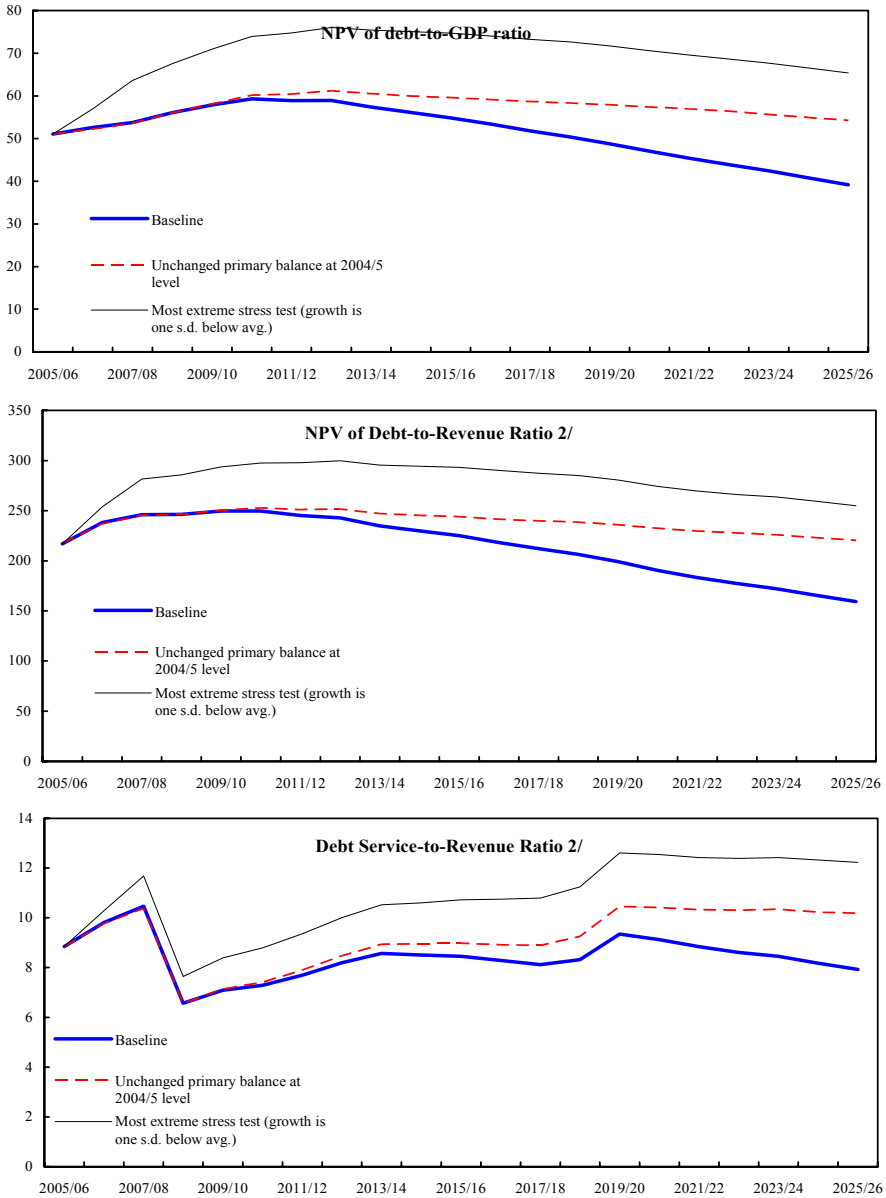
6/ External debt service is presented before debt relief until 2007 for comparability with the authorities' medium-term fiscal framework.

Table 2b.Ethiopia: Sensitivity Analysis for Key Indicators of Public Debt 2005/6-2025/26

	Estimate	Projections						
	2005/6	2006/7	2007/8	2008/9	2009/10	2010/11	2015/16	2025/26
NPV of Debt-to-GDP Ratio								
Baseline	51	53	54	56	58	59	55	39
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	51	53	54	57	60	62	63	58
A2. Primary balance is unchanged from 2004/5	51	52	54	56	58	60	60	54
A3. Permanently lower GDP growth 1/	51	53	56	59	62	65	70	77
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2006/7-2007/8	51	57	64	68	71	74	75	65
B2. Primary balance is at historical average minus one standard deviations in 2006/7-2007/8	51	54	57	59	60	62	57	41
B3. Combination of B1-B2 using one half standard deviation shocks	51	55	59	61	62	63	57	38
B4. One-time 30 percent real depreciation in 2006/7	51	60	61	64	66	68	62	44
B5. 10 percent of GDP increase in other debt-creating flows in 2006/7	51	60	60	62	63	64	59	42
NPV of Debt-to-Revenue Ratio 2/								
Baseline	217	238	246	246	250	250	225	159
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	218	239	249	251	256	259	253	230
A2. Primary balance is unchanged from 2004/5	217	237	245	246	250	253	244	220
A3. Permanently lower GDP growth 1/	217	241	253	256	264	269	275	290
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2006/7-2007/8	217	254	282	286	294	298	293	255
B2. Primary balance is at historical average minus one standard deviations in 2006/7-2007/8	217	246	262	258	260	259	235	167
B3. Combination of B1-B2 using one half standard deviation shocks	217	247	266	261	262	260	228	151
B4. One-time 30 percent real depreciation in 2006/7	217	274	280	281	285	286	254	181
B5. 10 percent of GDP increase in other debt-creating flows in 2006/7	217	274	275	270	271	269	241	169
Debt Service-to-Revenue Ratio 2/								
Baseline	9	10	10	7	7	7	8	8
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	9	10	10	7	7	8	9	11
A2. Primary balance is unchanged from 2004/5	9	10	10	7	7	7	9	10
A3. Permanently lower GDP growth 1/	9	10	11	7	8	8	10	13
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2006/7-2007/8	9	10	12	8	8	9	11	12
B2. Primary balance is at historical average minus one standard deviations in 2006/7-2007/8	9	10	11	7	8	8	9	8
B3. Combination of B1-B2 using one half standard deviation shocks	9	10	11	7	8	8	9	8
B4. One-time 30 percent real depreciation in 2006	9	10	11	7	8	8	9	9
B5. 10 percent of GDP increase in other debt-creating flows in 2006/7	9	10	12	8	8	8	9	9

Sources: Country authorities; and Fund staff estimates and projections. The fiscal year ends July 7.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of 20 (i.e., the length of the projection period).
2/ Revenues are defined inclusive of grants. Debt service ratios are presented before HIPC debt relief until 2007/8 for comparability with the authorities' medium-term fiscal framework. As a result, the ratios drop in 2008/9 once external debt service is expressed net of all debt relief.

Figure 3.Ethiopia: Indicators of Public Debt Under Alternative Scenarios, 2005/6-2025/26 1/



Source: Staff projections and simulations.
1/ Most extreme stress test is test that yields highest ratio in 2015/16.
2/ Revenues are defined inclusive of grants. Debt service ratios are presented gross of HIPC debt relief until 2007/8 for comparability with the authorities' medium-term fiscal framework. As a result, the ratios drop in 2008/9 once external debt service is expressed net of all debt relief.